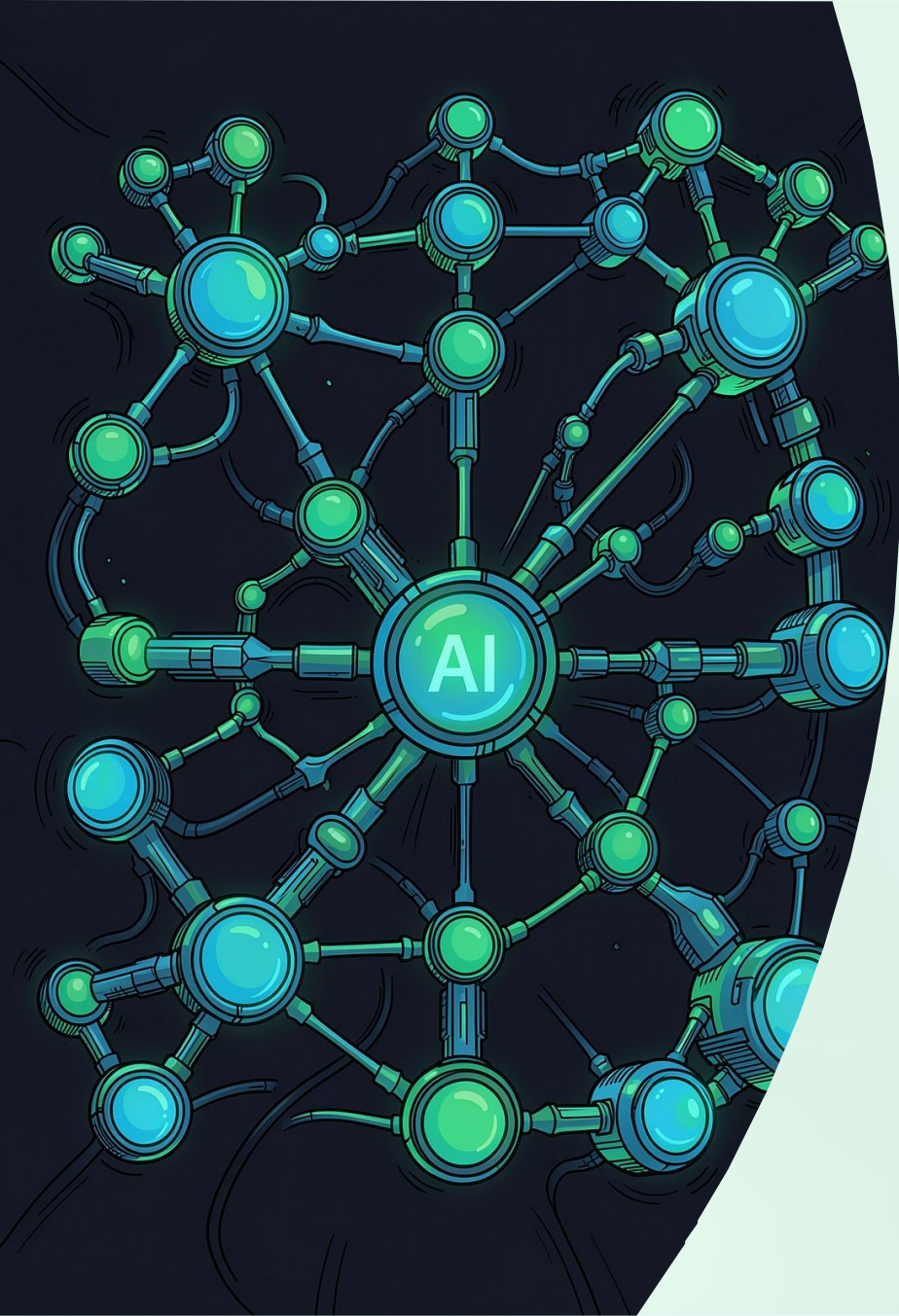




AI-Powered Event Management System

Using n8n and Generative AI

Automating the Complete Event Lifecycle using Workflow Automation and Artificial Intelligence

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The Challenge of Scaling Events

As participant numbers grow, manual event management becomes unsustainable — introducing errors, delays, and a poor experience for both organizers and attendees.

✗ Manual Approach

- Registration via forms & spreadsheets
- Confirmation emails sent by hand
- Physical attendance marking
- Reminder emails manually drafted
- Feedback reviewed one by one

✓ Automated Approach

- Form-triggered instant registration
- AI-generated personalized emails
- QR-based digital check-in
- Scheduled smart reminders
- AI sentiment analysis on feedback

Problem Statement

Traditional event management relies on repetitive manual processes that are error-prone, slow, and difficult to scale.

Human Errors

Manual data entry leads to duplicate records, typos, and missed participants.

Delayed Communication

Confirmation and reminder emails are sent late or forgotten entirely.

Time-Consuming Attendance

Physical check-ins create queues and consume organizer bandwidth.

Manual Feedback Analysis

Reading and summarizing hundreds of responses is impractical without AI.

📌 **Project Goal:** Build an end-to-end automated event management system powered by n8n workflows and Gemini AI — eliminating manual effort at every stage.

Project Objectives



Automated Registration

Trigger workflows on Google Form submission with instant ID generation.



AI Email Generation

Gemini AI drafts personalized welcome, reminder, and notification emails.



AI Feedback Analysis

Automated sentiment scoring and summary generation from feedback forms.



QR Code Generation

Auto-generate unique QR codes for every registered participant.



QR Attendance System

Scan-to-check-in via webhook with real-time Google Sheets updates.



Reduce Organizer Workload

Eliminate repetitive manual tasks so organizers focus on strategy.

Existing System vs. Proposed System

The proposed system replaces every manual touchpoint with intelligent automation — reducing errors, accelerating processing, and improving the participant experience at every stage.

Existing System	Proposed System
Manual Registration	Automated Registration
Manual Emails	AI-Generated Emails
Manual Attendance	QR-Based Attendance
Manual Feedback Analysis	AI Sentiment Analysis
High Human Error	Low Human Error
Time-Consuming	Fast Processing

Faster Processing

Instant responses instead of delayed manual actions

Low Error Rate

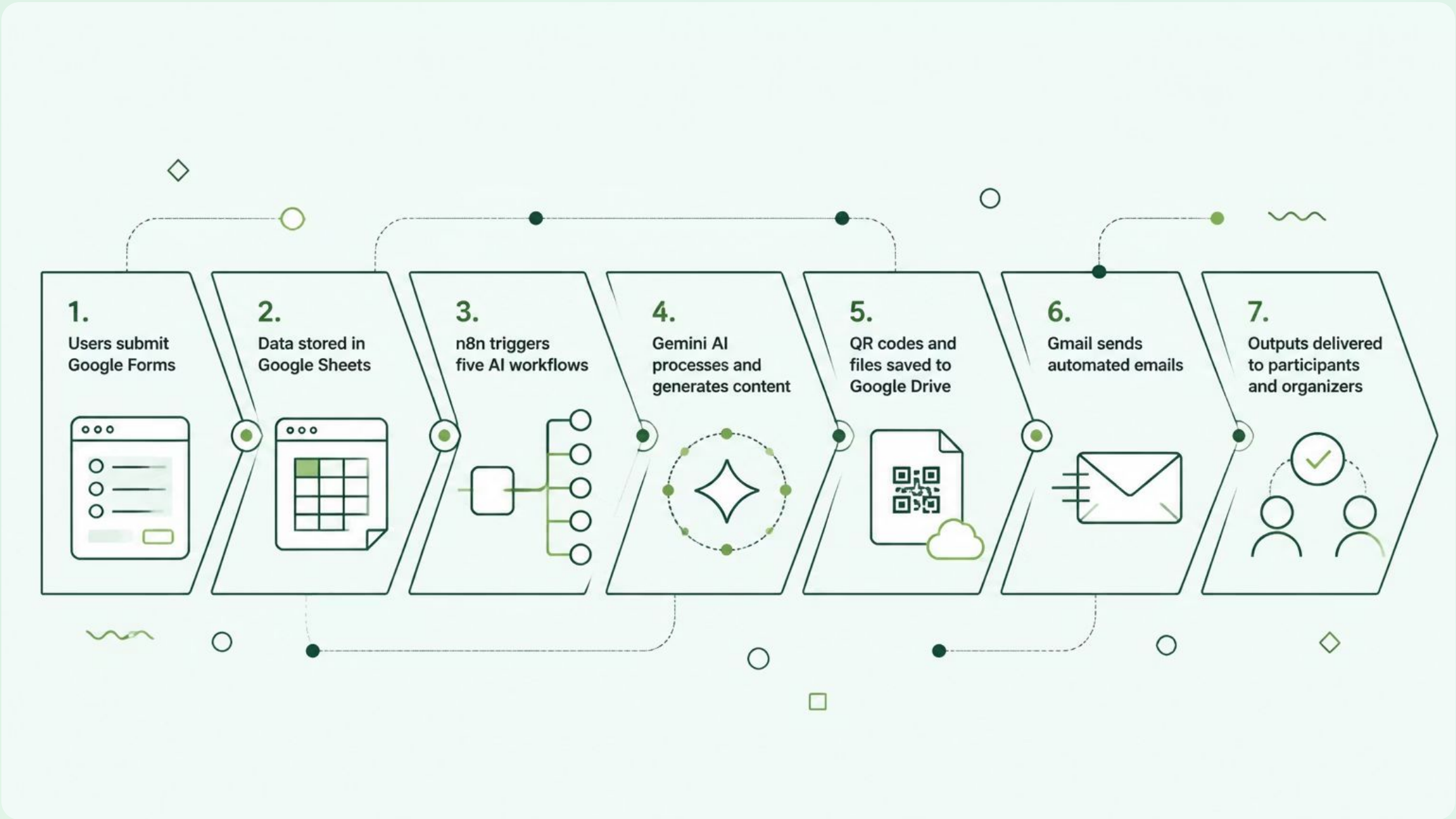
Automated validation eliminates duplicate records

AI-Powered

Gemini AI handles content generation and analysis

System Architecture

The system uses a layered, event-driven architecture where Google Workspace acts as the data layer, n8n orchestrates five intelligent workflows, and Gemini AI powers all content generation and analysis.



Input Layer
Google Forms & Sheets

Orchestration
n8n Workflows

AI Engine
Gemini AI

Output Layer
Gmail, Drive, Sheets

Core Workflows 1–3

N8N AUTOMATION

1

Registration & Participant Management

Form submission → Generate
Registration ID → Generate QR Code
→ Upload to Drive → AI Welcome
Email → Confirmation Email

2

QR Check-In System

Scan QR → Webhook Trigger → Verify
Participant → Update Attendance
Sheet → Instant Response

3

AI Reminder Email Automation

Schedule Trigger → Read Google
Sheets → Gemini AI drafts reminder
→ Send via Gmail

  **Workflow Screenshots:** n8n canvas for Workflows 1–3, Google Form, generated QR code, and sample AI email output.

Core Workflows 4–5

N8N AUTOMATION

1

AI Notification Automation

Read Participant Data → JavaScript Formatting → Gemini AI personalizes content → Send Notification Email via Gmail

2

AI Feedback Analyzer

Feedback Form → Google Sheets → Gemini AI Sentiment Analysis → Generate Summary → Update Sheet with scores

  **Workflow Screenshots:** n8n canvas for Workflows 4–5, sample notification email, and AI feedback sentiment summary in Google Sheets.

Technologies Used



n8n

Open-source workflow automation orchestrating all five AI pipelines.



JavaScript & Webhooks

Custom logic nodes and webhook triggers enable real-time event handling.



Gemini AI

Google's generative AI for email drafting, summarization, and sentiment analysis.



QuickChart API

Generates QR codes programmatically within the n8n registration workflow.



Google Workspace

Forms, Sheets, Drive, and Gmail serve as the data and communication backbone.

Results, Future Scope & Conclusion

Key Achievements

- End-to-end event automation
- Instant QR code generation
- AI-powered email & feedback analysis
- Reduced manual workload

Future Scope

- Certificate generation & WhatsApp integration
- Mobile app & dashboard analytics
- Payment gateway & SMS alerts
- Predictive analytics & multilingual support

Conclusion

This system demonstrates how n8n and Gemini AI can fully automate the event lifecycle — from registration to feedback — with minimal human intervention.

Thank You

Questions?

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